

The Boutet de Monvel conjecture for strongly convex Fourier multipliers on flat tori

Candidate partial result for arXiv:2110.09389
Run fa_banach_001, lane 10, model GPT5.6

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Status. Candidate partial result, likely valid, subject to expert review. The theorem below verifies all six conclusions of Conjecture 4.1 for a class of translation-invariant analytic elliptic operators on flat tori. The class contains fourth-order differential operators that are not powers of a Laplacian. The conjecture on general analytic manifolds remains open.

1 Source problem

The source is D. S. Winterrose, *Algebras of pseudo-differential operators acting on holomorphic Sobolev spaces*, arXiv:2110.09389, Conjecture 4.1 on printed page 8 [1]. For an analytic, formally self-adjoint, nonnegative elliptic polyhomogeneous pseudodifferential operator P of order $d > 0$, with positive principal symbol p and strictly convex unit co-balls, the conjecture asserts six conclusions. In abbreviated form:

- (1) the imaginary-time Hamilton flow of $p^{1/d}$ identifies $B_\varepsilon^*M \setminus 0$ with a complex tube $M_\varepsilon^\Phi \setminus M$;
- (2) the tube is open with real-analytic boundary;
- (3) it is strictly pseudoconvex;
- (4) the kernel of $e^{-\varepsilon P^{1/d}}$ extends holomorphically in its first variable to the tube;
- (5) its boundary value is a polyhomogeneous complex-phase Fourier integral operator of order $-(n-1)/4$;
- (6) that operator is a homeomorphism $H^s(M) \rightarrow \mathcal{O}^{s+(n-1)/4}(\partial M_\varepsilon^\Phi)$ for every $s \in \mathbb{R}$.

Here $\mathcal{O}^r(\partial D)$ is the closure in $H^r(\partial D)$ of smooth boundary values of functions holomorphic in D , as in the source. The full source statement is reproduced in Figure 1.

2 Statement of the partial result

Write $\mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$ and $\mathbb{T}_{\mathbb{C}}^n = (\mathbb{C}/2\pi\mathbb{Z})^n$. Let $q : \mathbb{R}^n \rightarrow [0, \infty)$ have the following properties:

- (Q1) $q(0) = 0$, while q is positive, even, positively one-homogeneous, and real analytic on $\mathbb{R}^n \setminus 0$;
- (Q2) for $F(\xi) = \frac{1}{2}q(\xi)^2$, the Hessian $D^2F(\xi)$ is positive definite for every $\xi \neq 0$.

Thus q is a strongly convex analytic Minkowski norm. Its dual norm is

$$q^*(v) = \sup_{q(\xi)=1} v \cdot \xi.$$

For $d > 0$, define the Fourier multiplier $P = P_{q,d}$ by

$$Pe_k = q(k)^d e_k, \quad e_k(x) = e^{ik \cdot x}, \quad k \in \mathbb{Z}^n,$$

with $q(0)^d = 0$. Equivalently, modulo a smoothing modification at the zero section, P has analytic classical symbol $p(\xi) = q(\xi)^d$.

Theorem 1 (All six conclusions on a flat torus). *Let q satisfy (Q1)–(Q2) and let $P = P_{q,d}$. Then for every $\varepsilon > 0$, all six conclusions of Winterrose’s Conjecture 4.1 hold. More precisely,*

$$D_\varepsilon := \{v \in \mathbb{R}^n : q^*(v) < \varepsilon\}, \quad M_\varepsilon^\Phi = \mathbb{T}^n + iD_\varepsilon,$$

and the flow diffeomorphism is

$$\Phi(x, \xi) = x + i\nabla F(\xi).$$

The boundary operator is

$$(S_\varepsilon f)(x + iv) = \sum_{k \in \mathbb{Z}^n} e^{-\varepsilon q(k)} \widehat{f}(k) e^{ik \cdot x - k \cdot v}, \quad q^*(v) = \varepsilon, \quad (1)$$

and it is a complex-phase polyhomogeneous FIO of order $-(n-1)/4$ satisfying

$$S_\varepsilon : H^s(\mathbb{T}^n) \xrightarrow{\sim} \mathcal{O}^{s+(n-1)/4}(\partial M_\varepsilon^\Phi) \quad (s \in \mathbb{R}).$$

In fact the admissible radii are unbounded: one may take $\varepsilon_0 = \varepsilon'_0 = +\infty$.

Corollary 1 (A non-Laplacian differential example). *There is $\delta_0 > 0$ such that, for every $0 < \delta < \delta_0$, the operator on $\mathbb{T}^2$*

$$P_\delta = (-\Delta)^2 + \delta \partial_1^2 \partial_2^2$$

satisfies all six conclusions of Conjecture 4.1 for every $\varepsilon > 0$. Its principal symbol

$$p_\delta(\xi) = (\xi_1^2 + \xi_2^2)^2 + \delta \xi_1^2 \xi_2^2$$

is not the square of a quadratic form when $\delta \neq 0$; hence this is not the flat-Laplacian case in disguise.

3 Proof intuition

On a translation-invariant torus the Hamiltonian covector never moves. At the imaginary time prescribed in the conjecture, its base point moves by $i\nabla(\frac{1}{2}q^2)$. Legendre duality identifies this gradient map exactly with the dual-norm tube $q^*(\text{Im } z) < \varepsilon$. The same duality gives the sharp estimate that makes the Fourier series of the Poisson kernel holomorphic in the tube.

On the boundary, Poisson summation turns the Fourier series into the positive complex phase

$$(x - y) \cdot \xi + i(\varepsilon q(\xi) + v \cdot \xi).$$

The imaginary part is nonnegative by duality and vanishes precisely at the unique support point $v = -\varepsilon \nabla q(\xi)$. Strong convexity makes this a nondegenerate positive phase. The standard order formula gives $-(n-1)/4$. Finally, Laplace’s method on the dual unit sphere shows that $S_\varepsilon^* S_\varepsilon$ is elliptic of order $-(n-1)/2$. This gives the exact Sobolev gain; Fourier polynomials and closed range then identify the range with the whole holomorphic Sobolev space.

The mechanism relies decisively on translation invariance. It does not prove the conjecture for variable symbols or curved manifolds.

4 Proof

4.1 Legendre geometry and the imaginary flow

Lemma 1 (Gradient parametrization of the dual ball). *Let $F = \frac{1}{2}q^2$ and let F^* be its Fenchel dual. Then*

$$F^*(v) = \frac{1}{2}q^*(v)^2.$$

Moreover, $\nabla F : \mathbb{R}^n \setminus 0 \rightarrow \mathbb{R}^n \setminus 0$ is a real-analytic diffeomorphism with inverse ∇F^ , and*

$$q^*(\nabla F(\xi)) = q(\xi).$$

Consequently ∇F maps $\{0 < q < \varepsilon\}$ real-analytically and diffeomorphically onto $\{0 < q^ < \varepsilon\}$.*

Proof. The positive Hessian and two-homogeneity of F make F strictly convex and coercive. Hence its gradient is injective and proper away from the origin, and the inverse-function theorem makes it a global real-analytic diffeomorphism of the punctured spaces. Equality in Fenchel duality gives $\nabla F^* = (\nabla F)^{-1}$. Maximizing $v \cdot \xi - \frac{1}{2}q(\xi)^2$ first along every ray gives $F^*(v) = \frac{1}{2}q^*(v)^2$.

Euler's identity gives $\xi \cdot \nabla F(\xi) = 2F(\xi) = q(\xi)^2$. Equality in Fenchel duality and one-homogeneity then give $q^*(\nabla F(\xi)) = q(\xi)$. The last assertion follows. $\square$

The Hamiltonian of the conjecture is $p^{1/d} = q$. Its flow is explicit:

$$\varphi_t(x, \xi) = (x + t\nabla q(\xi), \xi).$$

It is entire in t . At the time $t = iq(\xi)$ used in Conjecture 4.1,

$$\pi_x \varphi_{iq(\xi)}(x, \xi) = x + iq(\xi)\nabla q(\xi) = x + i\nabla F(\xi).$$

Lemma 1 therefore proves conclusion (1), with the asserted tube, for every $\varepsilon > 0$.

The function $F^* = \frac{1}{2}(q^*)^2$ is real analytic away from the origin, and its Hessian is positive definite there: at dual points it is the inverse of D^2F . Thus

$$r(x + iv) = 2F^*(v) - \varepsilon^2$$

is a real-analytic defining function near $\partial M_\varepsilon^\Phi$. Its complex Levi form is, up to a positive fixed factor, the restriction of $D^2F^*(v)$. It is positive definite at every boundary point. This proves conclusions (2) and (3).

4.2 Holomorphic continuation of the kernel

With normalized Haar measure, the kernel of $e^{-\varepsilon P^{1/d}}$ is

$$K_\varepsilon(z, y) = (2\pi)^{-n} \sum_{k \in \mathbb{Z}^n} e^{-\varepsilon q(k)} e^{ik \cdot (z-y)}. \quad (2)$$

Put $z = x + iv$. Duality gives

$$k \cdot v \geq -q(k)q^*(v),$$

and hence

$$\left| e^{-\varepsilon q(k)} e^{ik \cdot (z-y)} \right| \leq e^{-(\varepsilon - q^*(v))q(k)}.$$

On every compact subset of D_ε , the right side has uniform exponential decay because $q(k) \asymp |k|$. The same remains true after arbitrary derivatives, which only add polynomial factors in k . Thus (2) converges normally with all derivatives and is holomorphic in z . This proves conclusion (4).

4.3 The positive complex phase and its order

Let $\Sigma_\varepsilon = \partial D_\varepsilon$. After a smooth removal of the zero frequency, Poisson summation expresses the kernel locally, modulo a smooth kernel, as a finite relevant lattice translate of

$$(2\pi)^{-n} \int_{\mathbb{R}^n} e^{i(x-y) \cdot \xi - \varepsilon q(\xi) - v \cdot \xi} a(\xi) d\xi, \quad v \in \Sigma_\varepsilon, \quad (3)$$

where a is a classical symbol of order zero equal to one for large $|\xi|$. Equivalently, its complex phase is

$$\phi(x, v, y, \xi) = (x - y) \cdot \xi + i(\varepsilon q(\xi) + v \cdot \xi). \quad (4)$$

For $q^*(v) = \varepsilon$,

$$\text{Im } \phi = \varepsilon q(\xi) + v \cdot \xi \geq 0.$$

Strong convexity makes the support point unique, so equality holds exactly when

$$v = -\varepsilon \nabla q(\xi). \quad (5)$$

At such a point, $\partial_\xi \phi = 0$ also forces $x = y$ (locally modulo $2\pi\mathbb{Z}^n$). The mixed $(x - y, \xi)$ block is the identity, while strict convexity is the positive transverse Hessian condition. Hence (4) is a nondegenerate positive complex phase in the sense of Melin–Sjöstrand [4]. Its canonical relation is the graph

$$(y, \xi) \longmapsto (x = y, v = -\varepsilon \nabla q(\xi); \xi \cdot dx). \quad (6)$$

Indeed, ξ is normal to Σ_ε at the support point, so the $i\xi \cdot dv$ part restricts to zero on $T_v \Sigma_\varepsilon$.

There are $N = n$ phase variables, the target has dimension $2n - 1$, the source has dimension n , and the amplitude has order zero. The distributional FIO order is therefore

$$0 + \frac{N}{2} - \frac{(2n - 1) + n}{4} = -\frac{n - 1}{4}.$$

The amplitude and phase are classical homogeneous away from the zero section, so the operator is polyhomogeneous. This is conclusion (5), using the standard positive complex-phase FIO calculus [3, 4].

4.4 Ellipticity and the exact Sobolev shift

The order and ellipticity can also be seen directly from the Gram operator. Translation invariance diagonalizes $S_\varepsilon^* S_\varepsilon$ in the e_k basis. For any smooth positive density $b(v) d\sigma(v)$ on Σ_ε , its eigenvalue is

$$A_\varepsilon(k) = e^{-2\varepsilon q(k)} \int_{\Sigma_\varepsilon} e^{-2k \cdot v} b(v) d\sigma(v) > 0. \quad (7)$$

For $k = \lambda\theta$, $\lambda \rightarrow \infty$, the exponent has a unique maximum at $v = -\varepsilon \nabla q(\theta)$. Its tangential Hessian is negative definite by strong convexity. Uniform Laplace asymptotics on the compact direction sphere therefore give the full classical expansion

$$A_\varepsilon(k) \sim |k|^{-(n-1)/2} \left(a_0 \left(\frac{k}{|k|} \right) + \frac{a_1(k/|k|)}{|k|} + \dots \right), \quad a_0 > 0. \quad (8)$$

Thus $S_\varepsilon^* S_\varepsilon$ is elliptic of order $-(n - 1)/2$.

Choose a positive, translation-invariant elliptic operator Λ of order one on $\mathbb{T}^n \times \Sigma_\varepsilon$ to define the boundary Sobolev norms. The graph FIO composition calculus applied to

$$B_r = S_\varepsilon^* \Lambda^{2r} S_\varepsilon$$

shows that B_r is a positive elliptic pseudodifferential operator of order

$$2r - \frac{n-1}{2}.$$

The principal symbol is positive because the graph (6) is injective and $a_0 > 0$. With $r = s + (n-1)/4$, the order is $2s$. Since all Fourier modes have strictly positive norm under S_ε and the high-frequency symbol is uniformly elliptic, the finitely many low modes cause no loss. Consequently

$$c_s \|f\|_{H^s(\mathbb{T}^n)} \leq \|S_\varepsilon f\|_{H^{s+(n-1)/4}(\partial M_\varepsilon^\Phi)} \leq C_s \|f\|_{H^s(\mathbb{T}^n)}. \quad (9)$$

In particular the range is closed.

It remains only to identify the range. Every finite Fourier sum

$$u(z) = \sum_{k \in E} c_k e^{ik \cdot z}$$

is entire on $\mathbb{T}_\mathbb{C}^n$ and lies in the range, since the nonzero factor $e^{-\varepsilon q(k)}$ can be absorbed into its input coefficient. Conversely, if u is holomorphic in M_ε^Φ and smooth on its closure, its Fourier coefficients in x satisfy the Cauchy–Riemann equations

$$u_k(v) = c_k e^{-k \cdot v}.$$

Smoothness makes its Fourier truncations converge in every boundary Sobolev norm. Hence Fourier polynomials are dense in $\mathcal{O}^r(\partial M_\varepsilon^\Phi)$ by the definition of that space. The range of S_ε is both closed and contains this dense set, so it equals $\mathcal{O}^r$. Together with (9), this proves conclusion (6) and Theorem 1.

4.5 The explicit quartic family

For

$$p_\delta(\xi) = |\xi|^4 + \delta \xi_1^2 \xi_2^2, \quad q_\delta = p_\delta^{1/4},$$

the function $F_\delta = \frac{1}{2} p_\delta^{1/2}$ is two-homogeneous and real analytic away from zero. At $\delta = 0$, $D^2 F_0 = I$. Hessian positivity therefore persists for all sufficiently small $|\delta|$ by continuity on the compact unit circle and zero-homogeneity of the Hessian. Taking $0 < \delta < \delta_0$ also makes p_δ positive.

If p_δ were the square of a positive quadratic form $a\xi_1^2 + 2b\xi_1\xi_2 + c\xi_2^2$, the missing cubic monomials in its square would force $b = 0$. The pure quartic coefficients then force $a = c = 1$, and the mixed coefficient would be 2, contradicting the actual value $2 + \delta$. This proves Corollary 1.

5 Verification notes

1. **Hypotheses.** The symbol q^d is positive and homogeneous of order d away from zero. The corresponding torus multiplier is formally self-adjoint, elliptic, and nonnegative. Its unit co-ball is $\{q \leq 1\}$, which is strictly convex under (Q2). The value at $k = 0$ is a finite-rank, hence smoothing, issue.

2. **Signs.** On $z = x + iv$, $e^{ikz} = e^{ikx - kv}$. Therefore the boundary equality locus is the *minimizing* support point $v = -\varepsilon \nabla q(k)$, consistent in (4), (5), and (7).
3. **Order check.** The complex phase uses n variables on a kernel whose base dimension is $(2n - 1) + n$. This independently gives $n/2 - (3n - 1)/4 = -(n - 1)/4$. Squaring produces order $-(n - 1)/2$, matching the $|k|^{-(n-1)/2}$ Laplace asymptotic.
4. **Sobolev check.** Inserting a boundary operator of order $2r$ gives Gram order $2r - (n - 1)/2$. Setting $r = s + (n - 1)/4$ yields exactly $2s$, so no dimension shift is missing.
5. **No computational dependency.** No numerical experiment or computer algebra statement is used in the proof.

Verifier report. The argument was checked internally by two independent order calculations: the phase-dimension formula and the Gram/Laplace calculation. The flow formula, duality sign, boundary support point, and Sobolev shift agree across the two routes. No contradiction was found. The most important expert-review point is the invocation of the standard positive complex-phase FIO composition calculus between (4) and (9); the explicit Gram asymptotic supplies an independent ellipticity check but does not replace every microlocal bookkeeping detail for arbitrary real s .

6 Scope and bounded novelty check

This is a full proof of the conjecture only for the stated flat-torus Fourier multiplier class, so it is packaged as a partial result relative to the general problem. It does not treat x -dependent symbols, curved manifolds, lower-order terms that do not commute with the multiplier, or loss of strong convexity.

On 21 July 2026, bounded searches were made in the run’s registry, solution, attempt, and proof-gap indexes and on arXiv/web search using the phrases “Boutet de Monvel conjecture flat torus pseudodifferential strongly convex Fourier multiplier”, “Boutet de Monvel Conjecture torus”, “holomorphic Sobolev spaces anisotropic torus Poisson kernel”, and “imaginary time Hamiltonian flow Minkowski norm torus Poisson kernel”. No exact result or close arXiv treatment of this anisotropic flat-torus class was found. The source paper records only the Laplacian case as proved. This is evidence for novelty, not an exhaustive literature certification.

Human-review recommendation. Review as a candidate new partial theorem. Priority checks are: (i) that the analytic pseudodifferential convention permits the harmless smoothing choice near the zero section; (ii) the positive complex-phase nondegeneracy and PHG order in (3)–(6); and (iii) the graph-composition argument for all real Sobolev exponents. The Legendre geometry, holomorphic Fourier convergence, Laplace ellipticity, and explicit quartic example are elementary and separately checkable.

References

- [1] D. S. Winterrose, *Algebras of pseudo-differential operators acting on holomorphic Sobolev spaces*, arXiv:2110.09389 (2021), especially Conjecture 4.1.
- [2] L. Boutet de Monvel, *Convergence dans le domaine complexe des séries de fonctions propres*, Journées Équations aux dérivées partielles (1978).

- [3] L. Hörmander, *Fourier integral operators. I*, Acta Math. **127** (1971), 79–183.
- [4] A. Melin and J. Sjöstrand, *Fourier integral operators with complex-valued phase functions*, Lecture Notes in Math. **459**, Springer (1975).

4. The Boutet de Monvel Conjecture. (A theorem in a prototypical case.)
 A theorem was announced by Boutet de Monvel in an old 1978 conference paper [2], but only the case $\sqrt{-\Delta_g}$ in [2] was proved by Stenzel in [17] and Zelditch in [23]. Several possible approaches to proving it generally are discussed by Zelditch in [23]. To our knowledge, the general statement is not yet fully proven.

Let $d > 0$, and let $P \in \Psi_{\text{phg}}^d(M)$ be analytic with classical principal symbol p . Suppose that P is formally self-adjoint, elliptic, $p|_{T^*M \setminus 0} > 0$, and that

$$\{\xi \in T_x^*M \mid p(x, \xi) \leq 1\} \text{ is strictly convex for all } x \in M.$$

In that case, P is semi-lower bounded, and we assume it has no negative eigenvalues. A real-analytic complete Hamiltonian flow $\varphi_t : T^*M \setminus 0 \rightarrow T^*M \setminus 0$ arises from $p^{\frac{1}{d}}$. Given any $t \in \mathbb{R}$ and $(x, \xi) \in T^*M \setminus 0$, it satisfies

$$\varphi_t(x, s\xi) = s \cdot \varphi_t(x, \xi) \text{ for all } s > 0,$$

and $\varphi_t(x, \xi)$ extends holomorphically in t to $\{t \in \mathbb{C} \mid |t| < \epsilon\}$ for $\epsilon > 0$ small enough. The extension is then a map

$$\{t \in \mathbb{C} \mid |t| < \epsilon\} \rightarrow T_{1,0}^*M_{\mathbb{C}} \setminus 0 : t \mapsto \varphi_t(x, \xi),$$

where $T_{1,0}^*M_{\mathbb{C}}$ is the holomorphic cotangent bundle of $M_{\mathbb{C}}$.

CONJECTURE 4.1 (Boutet de Monvel [2]). The statement here is a bit different).
 Let P_{ϵ} denote the Schwartz kernel of $e^{-\epsilon P^{\frac{1}{d}}}$, and put

$$\Phi_x(\xi) = (\pi_x \varphi_{ip(x, \xi)^{\frac{1}{d}}})(x, \xi) \text{ for any } (x, \xi) \in B_{\epsilon}^*M \setminus 0,$$

where π_x is the bundle projection, and

$$B_{\epsilon}^*M = \{(x, \xi) \in T^*M \mid p(x, \xi)^{\frac{1}{d}} < \epsilon\}.$$

There is a maximal $\epsilon_0 > 0$ such that for any $\epsilon \in (0, \epsilon_0)$ the following holds:

1. The extended flows combine into a real-analytic diffeomorphism

$$\Phi : B_{\epsilon}^*M \setminus 0 \rightarrow M_{\epsilon}^{\Phi} \setminus M \subset M_{\mathbb{C}} : (x, \xi) \mapsto \Phi_x(\xi).$$

2. The set M_{ϵ}^{Φ} is open in $M_{\mathbb{C}}$ with (orientable) C^{ω} -boundary $\partial M_{\epsilon}^{\Phi}$.

3. The set M_{ϵ}^{Φ} is strictly pseudo-convex.

(Point 3. implies that M_{ϵ}^{Φ} admits a Kähler metric h , and a global potential for it.)
 There is a maximal $\epsilon'_0 \in (0, \epsilon_0]$ such that for any $\epsilon \in (0, \epsilon'_0)$ the following holds:

4. The map $x \mapsto P_{\epsilon}(x, y)$ extends holomorphically to M_{ϵ}^{Φ} for each fixed $y \in M$.
5. The kernel $P_{\epsilon}|_{\partial M_{\epsilon}^{\Phi} \times M}$ induces a complex-phase PHG FIO S_{ϵ} of order $-\frac{n-1}{4}$.
6. S_{ϵ} defines a homeomorphism $S_{\epsilon} : H^s(M) \rightarrow \mathcal{O}^{s+\frac{n-1}{4}}(\partial M_{\epsilon}^{\Phi})$ for any $s \in \mathbb{R}$.

The first three points simply generalize the basic properties of the Grauert tubes. In the special case that $P = -\Delta_g$, M_{ϵ}^{Φ} is exactly the Grauert tube M_{ϵ} .

Figure 1: Conjecture 4.1 in arXiv:2110.09389, printed page 8.